

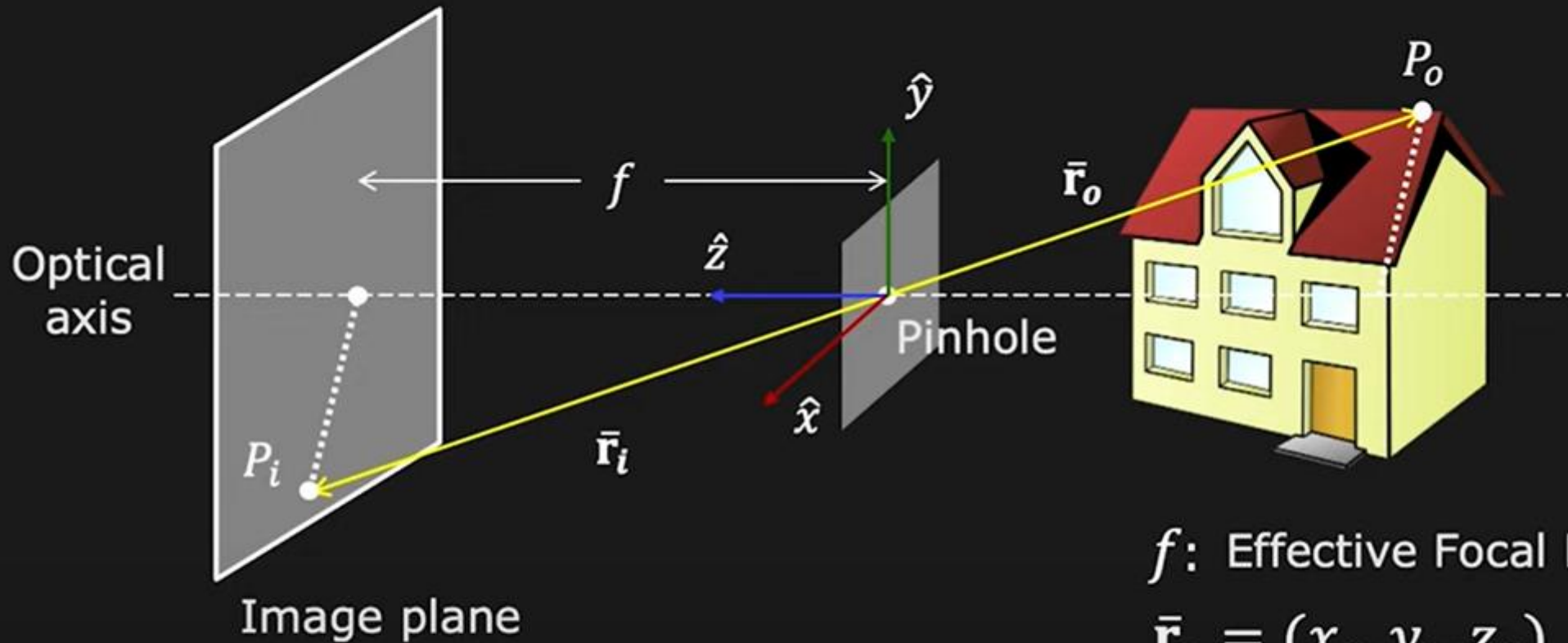
Image Formation

Image: Projection of 3D scene onto 2D plane. We need to understand the geometric and photometric relation between the scene and its image.

Topics:

- (1) Pinhole and Perspective Projection
- (2) Image formation using Lenses
- (3) Lens Related Issues
- (4) Wide Angle Cameras

Perspective Imaging with Pinhole



f : Effective Focal Length

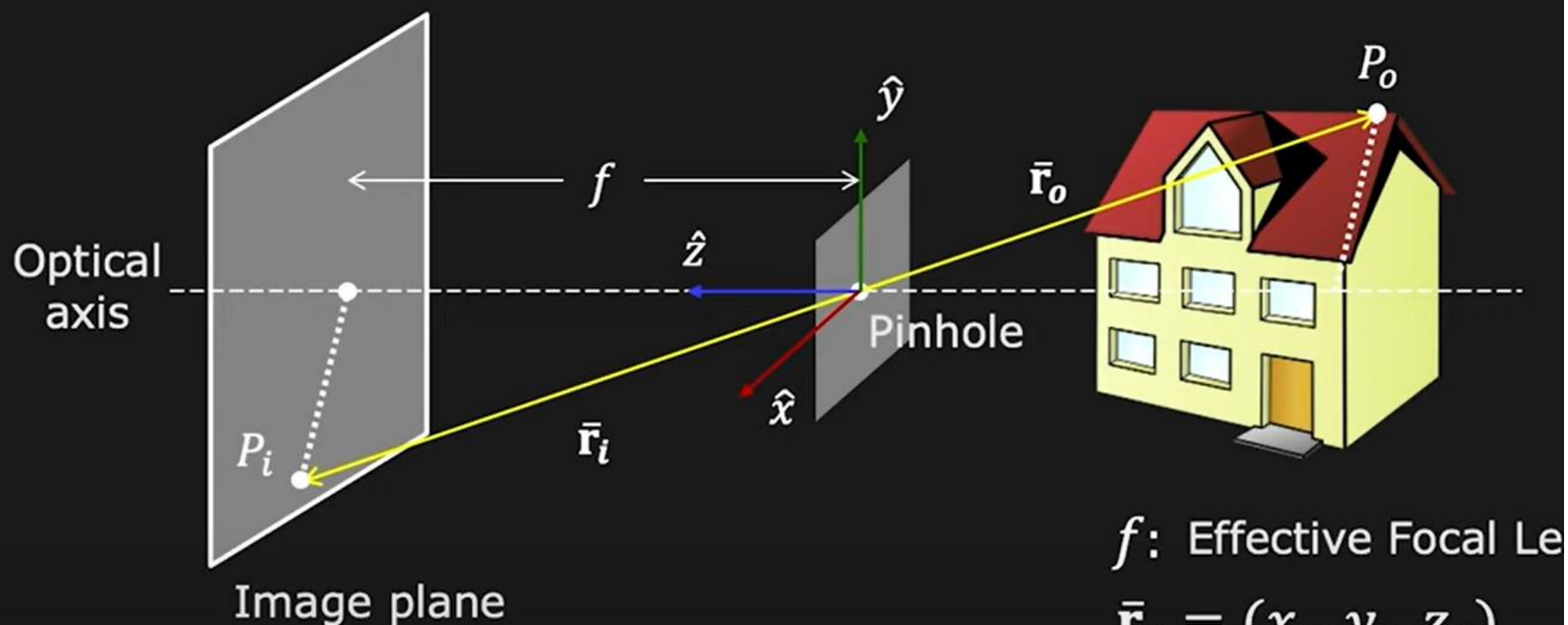
$$\bar{\mathbf{r}}_o = (x_o, y_o, z_o)$$

$$\bar{\mathbf{r}}_i = (x_i, y_i, f)$$

Using similar triangles:

$$\frac{\bar{\mathbf{r}}_i}{f} = \frac{\bar{\mathbf{r}}_o}{z_o}$$

Perspective Imaging with Pinhole



f : Effective Focal Length

$$\bar{\mathbf{r}}_o = (x_o, y_o, z_o)$$

$$\bar{\mathbf{r}}_i = (x_i, y_i, f)$$

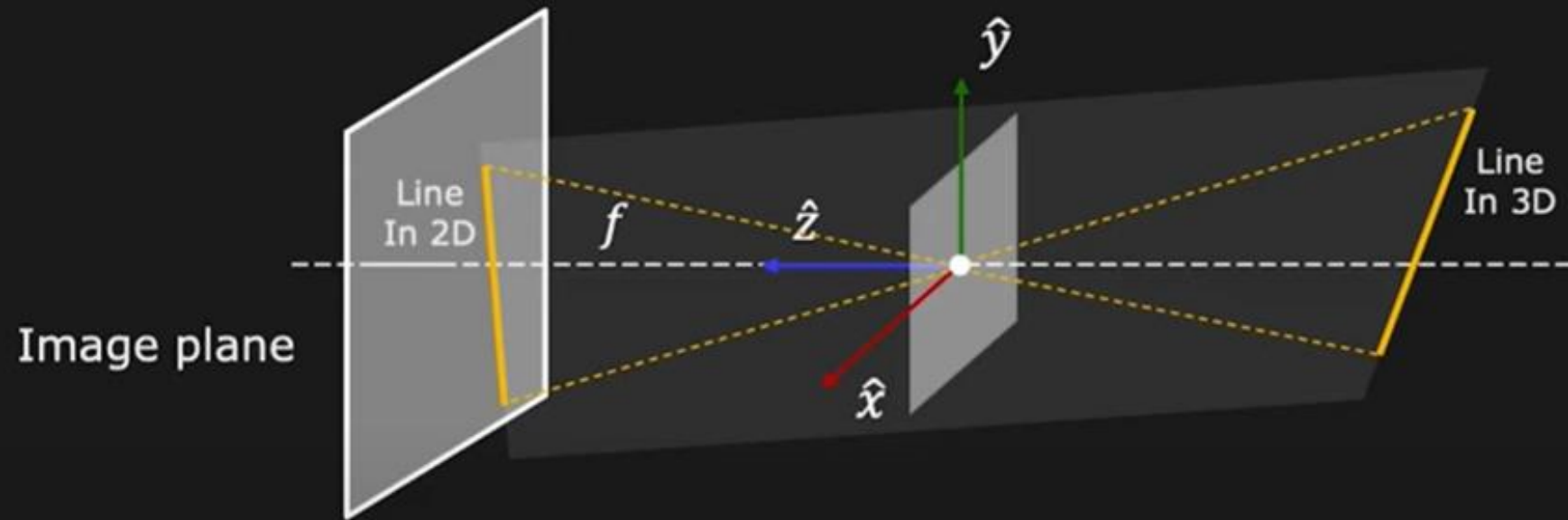
Using similar triangles:

$$\frac{\bar{\mathbf{r}}_i}{f} = \frac{\bar{\mathbf{r}}_o}{z_o}$$



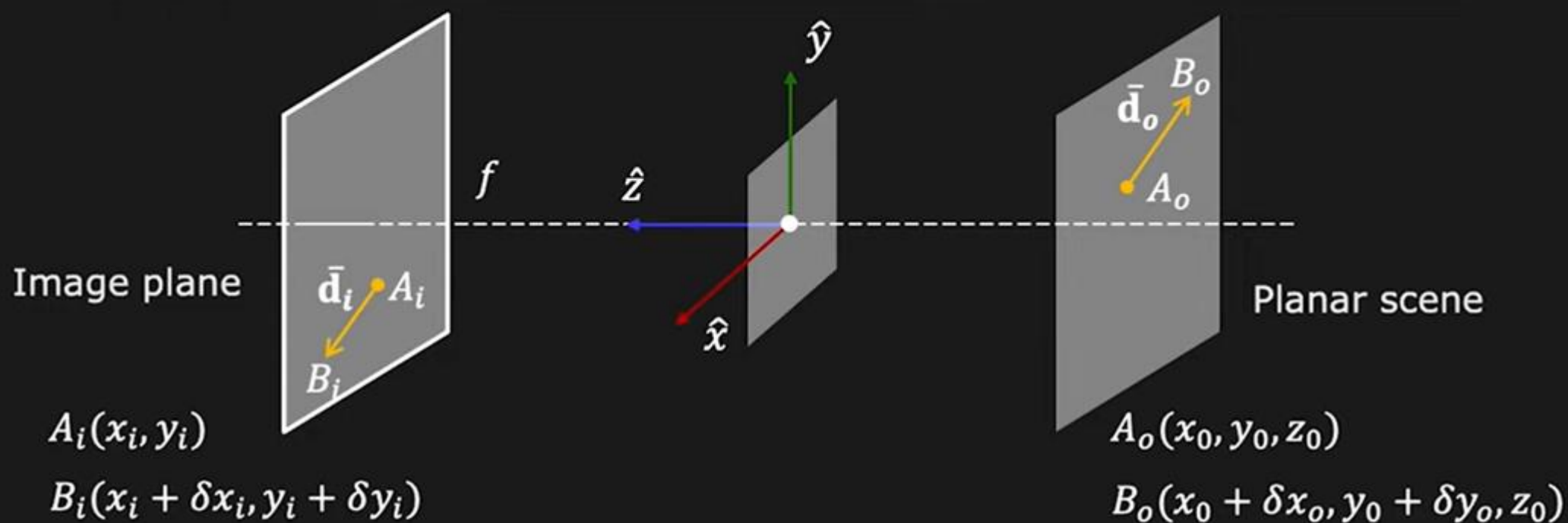
$$\frac{x_i}{f} = \frac{x_o}{z_o}, \quad \frac{y_i}{f} = \frac{y_o}{z_o}$$

Perspective Projection of a Line



Straight line in scene remains straight in image

Image Magnification



From Perspective Projection:

$$\frac{x_i}{f} = \frac{x_o}{z_o} \quad \text{and} \quad \frac{y_i}{f} = \frac{y_o}{z_o} \quad \text{----- (A)}$$

$$\frac{x_i + \delta x_i}{f} = \frac{x_o + \delta x_o}{z_o} \quad \text{and} \quad \frac{y_i + \delta y_i}{f} = \frac{y_o + \delta y_o}{z_o} \quad \text{----- (B)}$$

Image Magnification

From (A) and (B) we get:

$$\frac{\delta x_i}{f} = \frac{\delta x_o}{z_o} \quad \text{and} \quad \frac{\delta y_i}{f} = \frac{\delta y_o}{z_o}$$

Magnification:

$$|m| = \frac{\|\bar{\mathbf{d}}_i\|}{\|\bar{\mathbf{d}}_o\|} = \frac{\sqrt{\delta x_i^2 + \delta y_i^2}}{\sqrt{\delta x_o^2 + \delta y_o^2}} = \left| \frac{f}{z_o} \right|$$

$$m = \frac{f}{z_o}$$

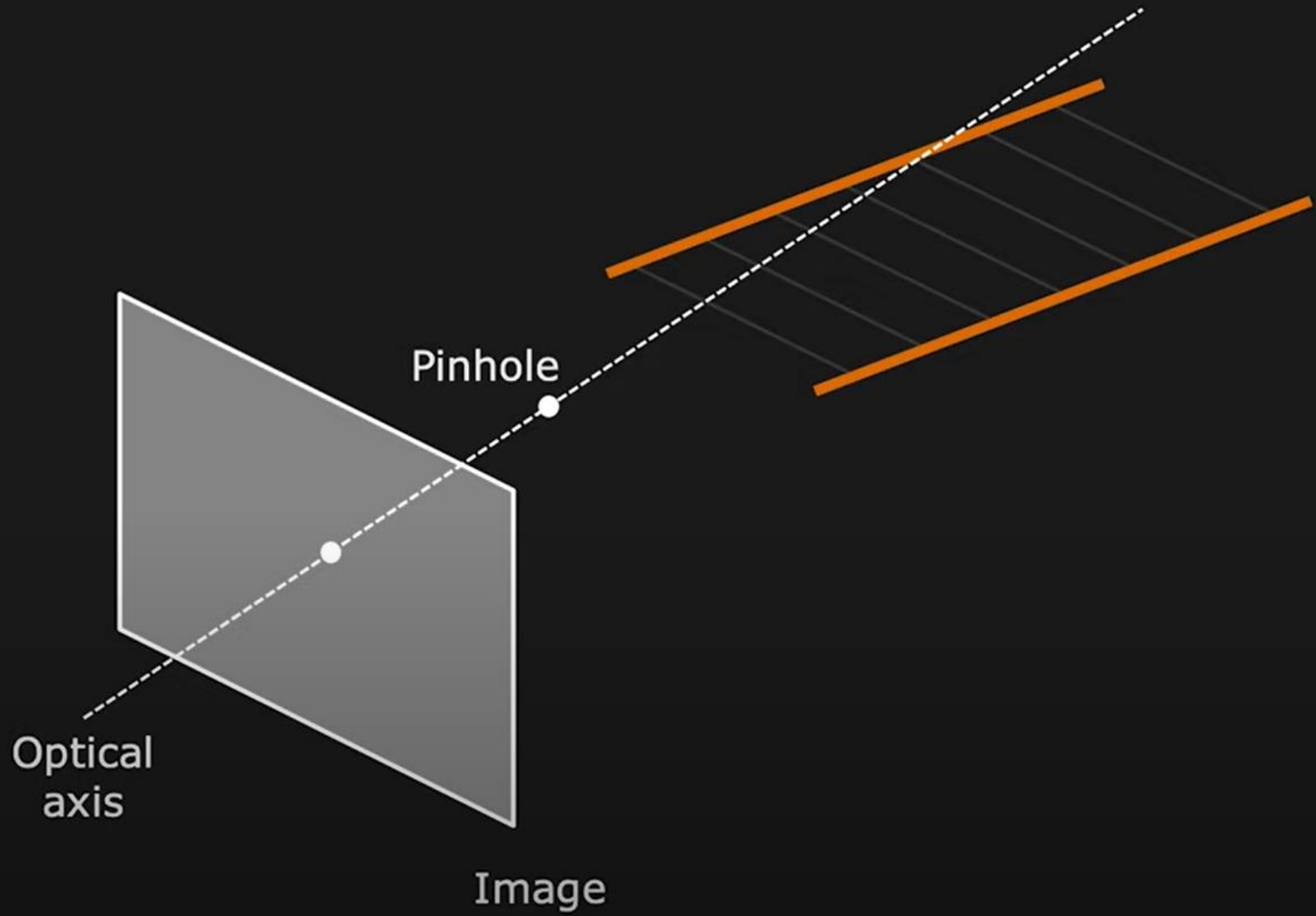
m is negative when image is **inverted**

Vanishing Point

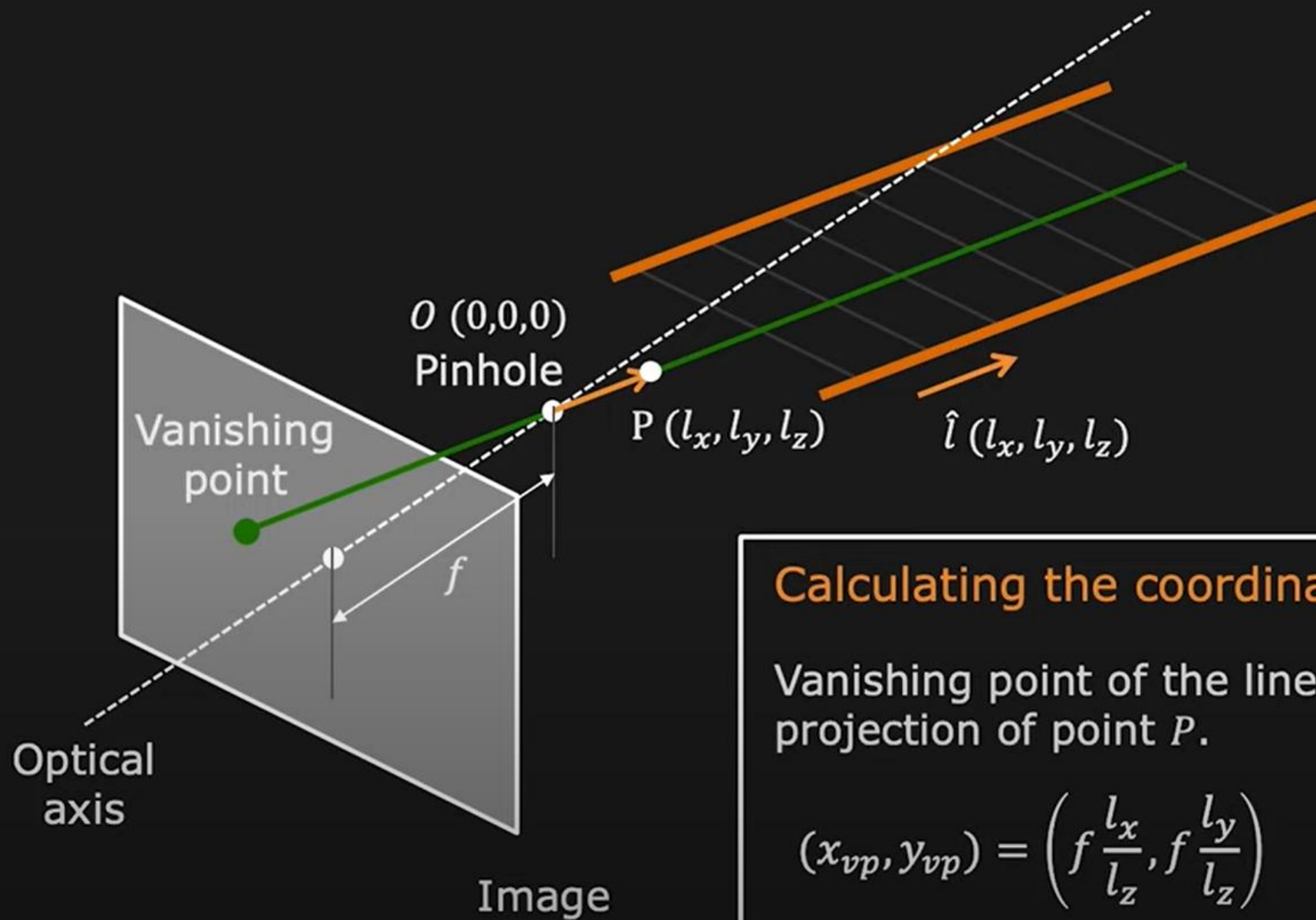


Parallel straight lines converge at a single image point

Finding the Vanishing Point



Finding Vanishing Point



Calculating the coordinates:

Vanishing point of the line is the projection of point P .

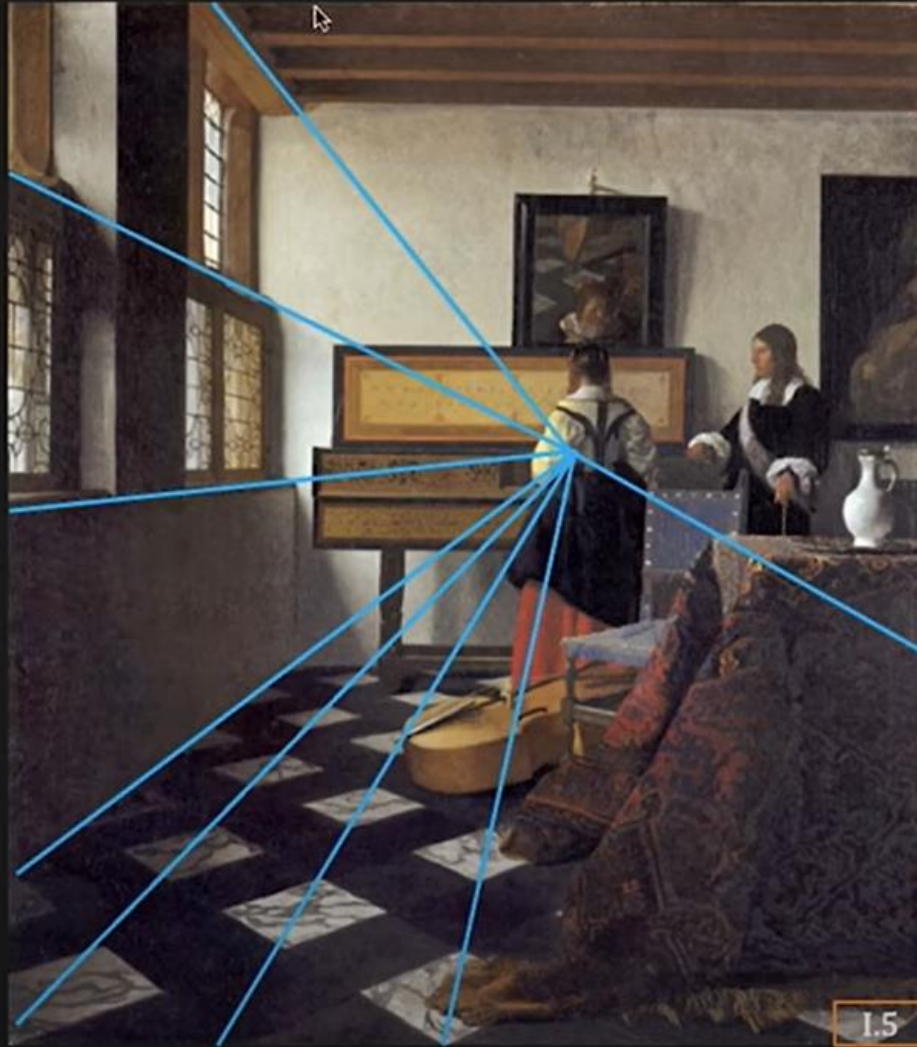
$$(x_{vp}, y_{vp}) = \left(f \frac{l_x}{l_z}, f \frac{l_y}{l_z} \right)$$

Use of Vanishing Point in Art



The Music Lesson, Johannes Vermeer, c. 1662-1664

Use of Vanishing Point in Art

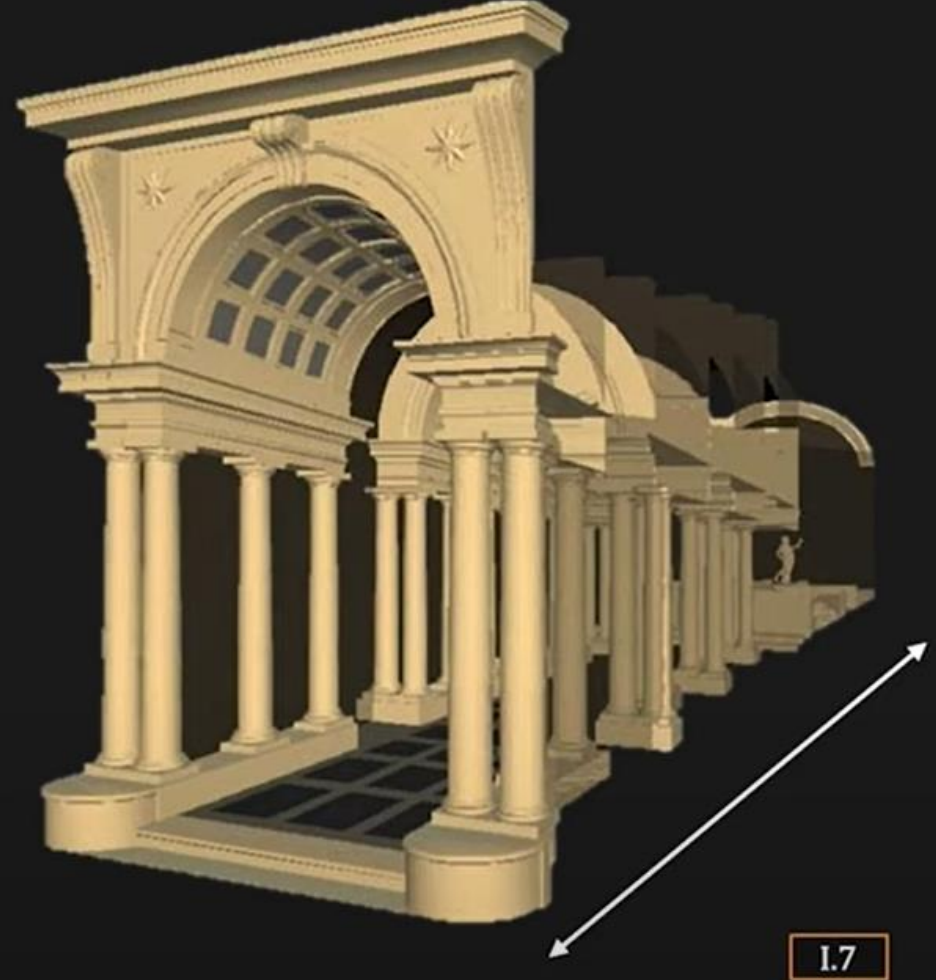


The Music Lesson, Johannes Vermeer, c. 1662-1664

False Perspective



Depth appears to be ~155 feet



Depth is actually ~30 feet